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BDNF/TrkB signaling regulates HNK-1 carbohydrate expression in regenerating motor nerves and promotes functional recovery after peripheral nerve repair

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Abstract

Functional recovery after peripheral nerve injury is often poor despite high regenerative capacity of peripheral neurons. In search for novel treatments, brief electrical stimulation of the acutely lesioned nerve has recently been identified as a clinically feasible approach increasing precision of axonal regrowth. The effects of this stimulation appear to be mediated by BDNF and its receptor, TrkB, but the down-stream effectors are unknown. A potential candidate is the HNK-1 carbohydrate known to be selectively reexpressed in motor but not sensory nerve branches of the mouse femoral nerve and to enhance growth of motor but not sensory axons in vitro. Here, we show that short-term low-frequency electrical stimulation (1 h, 20 Hz) of the lesioned and surgically repaired femoral nerve in wild-type mice causes a motor nerve-specific enhancement of HNK-1 expression correlating with previously reported acceleration of muscle reinnervation. Such enhanced HNK-1 expression was not observed after electrical stimulation in heterozygous BDNF or TrkB-deficient mice. Accordingly, the degree of proper reinnervation of the quadriceps muscle, as indicated by retrograde labeling of motoneurons, was reduced in TrkB+/− mice compared to wild-type littermates. Also, recovery of quadriceps muscle function, evaluated by a novel single-frame motion analysis approach, and axonal regrowth into the distal nerve stump, assessed morphologically, were considerably delayed in TrkB+/− mice. These findings indicate that BDNF/TrkB signaling is important for functional recovery after nerve repair and suggest that up-regulation of the HNK-1 glycan is linked to this phenomenon.

Keywords: BDNF; Femoral nerve; HNK-1 carbohydrate; Motion analysis; Motoneuron; Mouse; Regeneration; TrkB

Introduction

Injured peripheral neurons in mammals, in contrast to nerve cells in the central nervous system, can regenerate their axons over long distances. Despite this potential, functional recovery after surgical repair is often poor (Fu and Gordon, 1997; Lundborg, 2003). Choice of appropriate pathways by regenerating axons is considered to be a major determinant of successful functional recovery. A favorable model system to study appropriate reinnervation of denervated targets is the femoral nerve which bifurcates into two branches, the quadriceps motor nerve and the saphenous cutaneous branch (Fig. 1). After transection of this nerve proximal to the bifurcation, motor neurons preferentially reinnervate the quadriceps branch, when given equal access to both branches (Brushart, 1988, 1993; Brushart and Seiler, 1987; Brushart et al., 1998; Madison et al., 1996; Robinson and Madison, 2003, 2005).

In search of the cellular and molecular determinants underlying preferential motor reinnervation, we previously found that an unusual acidic glycan recognized by the monoclonal antibody HNK-1 is associated with myelin profiles of motor axons, but not of sensory axons in the mouse (Martini and Schachner, 1986; Martini et al., 1992, 1994). During development, HNK-1 is expressed by all premyelinating or non-myelinating Schwann cells and is down-regulated during early postnatal periods in these cells to then become up-

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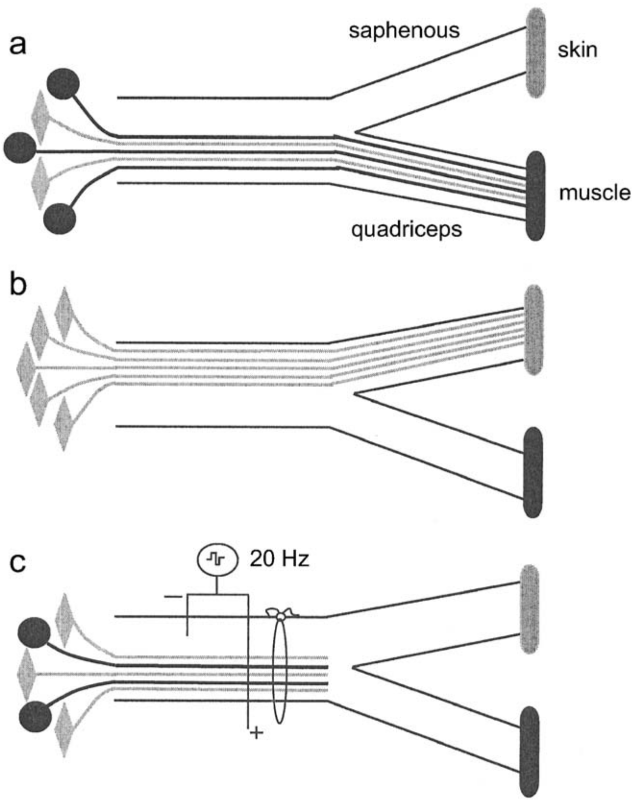

Fig. 1. Diagrammatic representation of the femoral nerve with its quadriceps (a) and saphenous branches (b). In intact animals, the quadriceps branch contains axons of motoneurons (black circles) innervating the quadriceps muscle as well as axons of sensory neurons (gray rhombuses) innervating muscle spindles and tendon organs. The saphenous branch is populated by axons of sensory neurons innervating the skin. The position of the bipolar electrode used to stimulate the transected nerve is shown in panel c.

regulated again in the myelin sheaths associated exclusively with motor axons. The HNK-1 carbohydrate is carried by or binds to different neural recognition molecules including the myelin associated glycoprotein MAG (Kruse et al., 1985), highly acidic glycolipids, several laminin isoforms (Hall et al., 1993), amphoterin (Mohan et al., 1992), the neural cell adhesion molecule (NCAM) and P0 (for reviews, see Kleene and Schachner, 2004; Schachner and Martini, 1995). These observations led to the hypothesis that the HNK-1 carbohydrate, which is detectable in compact myelin, basal lamina and at the cell surface of Schwann cells, the sites that motor axons follow as they regenerate after peripheral nerve damage, could be specifically involved in motor axon regrowth. This view was supported by in vitro experiments showing that substrate-coated HNK-1 carbohydrate enhances neurite outgrowth from motoneurons, but not from sensory neurons (Martini et al., 1992).

Previous work has shown that continuous low-frequency (20 Hz) electrical stimulation of the proximal nerve stump, performed immediately after nerve transection for a period as short as 1 h, dramatically reduces the period of protracted axonal regrowth and increases precision of reinnervation in the rat (Al Majed et al., 2000a). The accelerated preferential motor reinnervation after stimulation is associated with an accelerated and enhanced up-regulation of the neurotrophin BDNF and its cognate receptor TrkB in motor neurons (Al Majed et al., 2000b, 2004). Here, we tested the hypothesis that TrkB signaling exerts its effects on preferential motor reinnervation via modulation of HNK-1 epitope expression. The results described here show that TrkB signaling is important for functional recovery after injury and suggest that TrkB-dependent HNK-1 expression mediates these beneficial functions.

Materials and methods

Animals

To investigate the effect of electrical stimulation on HNK-1 expression, femoral nerve repair was performed in 72 C57BL/6J mice obtained from the breeding unit of the University Hospital Hamburg. To define the contribution of BDNF to this process, we used 18 heterozygous BDNF-deficient mice (Korte et al., 1995) kindly provided by Martin Korte (Max-Planck-Institute for Neurobiology, Martinsried). Sixteen wild-type littermate mice were used as controls for these experiments. Further experiments were performed on 28 heterozygous TrkB-deficient mice kindly provided by Louis F. Reichardt (University of California, San Francisco). Sixteen wild-type littermates served as controls. Half of the animals mentioned above were subjected to electrical stimulation after transection of the femoral nerve (see below). Additional 16 TrkB-deficient mice and 16 wild-type littermates were used for analysis of functional recovery after nerve repair and for subsequent morphological analyses of retrogradely labeled motoneurons and regenerated nerves. None of the animals in this group was subjected to electrical stimulation of the femoral nerve.

Breeding and genotyping of animals

Heterozygous BDNF and wild-type mice on a C57BL/6J background were bred in the animal facility of the University Hospital Hamburg and genotyped by standard polymerase chain reaction (PCR) assay at the Max-Planck-Institute for Neurobiology, Martinsried (Korte et al., 1995). Heterozygous TrkB mutants and wild-type littermates (C57BL/6J genetic background) were derived from heterozygous breeding and genotyped by PCR as described previously (Xu et al., 2000a,b).

Experimental paradigm

In the intact quadriceps branch of the femoral nerve, sensory and motor fibers are intermingled to innervate the skeletal muscles (Fig. 1a). The saphenous branch contains only sensory fibers that innervate the skin (Fig. 1b). Although after nerve injury motor axons preferentially reinnervate muscle pathways, many motor axons fail to find the appropriate muscle branch and regrow into the saphenous branch (Martini et al., 1992, 1994; Brushart et al., 1998; Robinson and Madison, 2005). Electrical stimulation of the proximal nerve stump after femoral nerve transection and surgical repair (Fig. 1c) causes accelerated regrowth of larger numbers of motor axons, as compared to sham-stimulation, into the appropriate motor nerve branch (Al Majed et al., 2000a).

Nerve repair

Experiments were performed under antiseptic conditions on femoral nerves of adult mice anesthetized with sodium pentobarbital (Somnotol, MTC Pharmaceuticals, Cambridge, Ontario, Canada, 30 mg/kg) or with fentanyl (Fentanyl-Janssen, Janssen, Neuss, Germany, 0.4 mg/kg), droperidol (Dehydrobenzperidol, Janssen, 20 mg/kg) and diazepam (Valium 10 Roche, Hoffman-La Roche, Grenzach-Wyhlen, Germany, 5 mg/kg) by intraperitoneal injection. The proximal femoral nerve was cut 3–4 mm proximal to the bifurcation into saphenous and quadriceps nerve branches. The proximal and distal stumps were inserted into a 4-mm-long silastic nerve cuff (0.37 mm inner diameter; Dow Corning, Midland, Michigan, USA). Using single stitches of 11-0 Nylon suture thread (Ethicon, Peterborough, ON, Canada) through the epineurium, the stumps were fixed so that their ends were separated by a gap of 2 mm. Surgery was performed unilaterally in all animals. Experiments were either approved by a local ethical committee (Health Science Laboratory Animal Services) under the Canadian guidelines for animal experimentation or were approved by the State of Hamburg animal care committee and conformed to EC and NIH guidelines.

Electrical stimulation

In experiments in which transected and repaired femoral nerves were electrically stimulated, two insulated Teflon-coated stainless steel wires (A5632; Cooner Wire Company, Chatsworth, CA) were bared of insulation for 2–3 mm, and each was twisted to form a small loop to secure on either side of the nerve stump proximal to the suture site (Fig. 1c). The cathode was sutured alongside the femoral nerve just below its exit from the peritoneal cavity, whereas the anode was sutured to the muscle close to the nerve, just proximal to the suture repair site. The wires were connected to a SD9 Square Pulse Stimulator (Grass Instrument Co., Quincy, MA, USA) to commence stimulation (20 Hz, single square pulses of 0.1 ms duration and 3 V amplitude) immediately. We chose a low stimulus frequency because it is the physiologically relevant frequency of hind limb motoneuron discharge (Loeb et al., 1987). A stimulation period of 1 h was selected in view of the equal effectiveness of 1 h stimulation and up to 2-week stimulation periods in accelerating axonal regeneration and preferential motor reinnervation (Al Majed et al., 2000a). In the sham-stimulated group of mice, the electrodes were implanted, but the stimulator was not switched on.

Immunohistochemistry

After survival time-periods of 1, 2 or 3 weeks, the animals were sacrificed by an overdose of Somnotol. The nerves were quickly removed, embedded in Tissue Tec® (Sakura, Tokyo, Japan) and frozen in liquid nitrogen as previously described by Bartsch et al. (1989). The embedded femoral nerves were cut into 10-μm thick cross-sections on a cryostat (CM3050, Leica). The sections were dried overnight, washed three times in phosphate-buffered saline (PBS, pH 7.4) for 10 min each, blocked with 2% bovine serum albumin (BSA, Sigma, Deisenhofen, Germany) for 2 h, washed again three times for 10 min each in PBS and incubated for 1 h with the rat monoclonal antibody 412 raised against the HNK-1 carbohydrate (Kruse et al., 1984, dilution 1:100). After incubation, the sections were washed in PBS, incubated for 1 h in PBS containing the secondary fluorescent antibody (Cy3-conjugated anti-rat IgG, Dianova, Hamburg, Germany; 1:500), washed again in PBS and coverslipped with Aqua Polymount (Polysciences, Warrington, PA).

Analysis of HNK-1 carbohydrate expression

Data were collected by two independent procedures. In one procedure, the number of HNK-1 positive profiles was evaluated using images obtained with a laser scanning microscope (LSM 510, Zeiss, 40× objective). Number of HNK-1 positive profiles per nerve cross-section was evaluated using confocal images (LSM 510 microscope, Zeiss, 40× objective) and an image processing program (Analysis, Zeiss). At least 5 sections cut distal from the bifurcation per nerve branch were analyzed. In the same sample, average relative fluorescence intensity was measured using the intensity measuring module of the microscope.

Analysis of motor function

Functional recovery was assessed by a novel single-frame motion analysis (SFMA) approach recently developed in our laboratory (Irintchev et al., 2005). To evaluate quadriceps muscle function during ground locomotion prior to operation, mice were trained to perform a classical beam walking test. In this test, the animals walk unforced from one end of a horizontal beam (length 1000 mm, width 38 mm) towards their home cage located at the other end of the beam. For all mice, a rear view of one walking trial was captured prior to the operation with a Panasonic NV-DS12 camera (Panasonic Deutschland, Hamburg, Germany) at 25 frames per second and recorded on videotape (video recorder SVL-SE 830, Sony Deutschland, Cologne, Germany). The recordings were repeated 1, 2, 4, 8 and 12 weeks after nerve transection. The video sequences were digitized and examined with VirtualDub software (available at http://www.virtualdub.org). Selected frames in which the animals were seen in defined phases of the step cycle (see below) were used for measurements performed with UTHSCSA ImageTool 2.0 software (University of Texas, San Antonia, TX, USA, http://ddsdx.uthscsa.edu/dig/). Two parameters were measured: the foot-base angle (FBA, Figs. 2a, b) and the heels-tail angle (HTA, Figs. 2c, d). The foot-base angle, measured at toe-off position, is defined by a line dividing the sole surface into two halves and the horizontal line, and is measured with respect to the medial aspect. The heels-tail angle is measured when one leg is in the single support phase and the contralateral

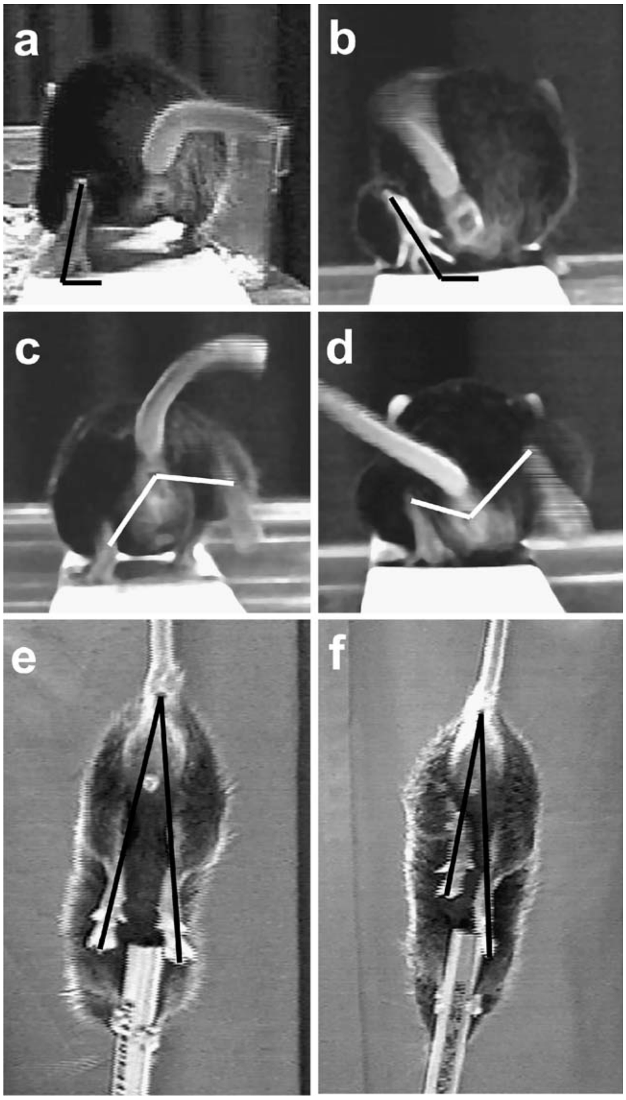

Fig. 2. Single video frames from recordings of beam walking (a–d) and limb protractions (e, f). Shown are intact animals (a, c, e) and animals with transection of the left femoral nerve at 7 days after lesion (b, d, f). The lines drawn in the video frames show the foot-base angle (FBA, a, b), the heels-tail angle (HTA, b, d) and the limb lengths used for calculation of the limb protraction length ratio (PLR, e, f). Note that HTA is measured always with respect to the dorsal aspect, FBA with respect to the medial aspect. For further explanations, see the main text.

extremity has maximum swing altitude, and is defined by the lines connecting the heels with the anus. The angle is measured with respect to the dorsal aspect. Compared to intact animals (Figs. 2a, c), the two angles change significantly after lesion of the femoral nerve (Figs. 2b, d). Both parameters are directly related to the ability of the quadriceps muscle to keep the knee joint extended during contralateral swing phases (see Irintchev et al., 2005). As a relative measure of functional recovery at 4 and 12 weeks after lesion, we calculated the stance recovery index which is a mean of the recovery index (RI) for the HTA and the FBA. The index is calculated, in percent, as:

$$\mathrm{RI} = [(X_{\mathrm{reinn}} - X_{\mathrm{den}})/(X_{\mathrm{pre}} - X_{\mathrm{den}})] \times 100,$$

where X_{pre}, X_{den} and X_{reinn} are values prior to operation, during the state of denervation (7 days after injury), and at 4 or 12 weeks of reinnervation, respectively.

A third parameter, the limb protraction length ratio (PLR), was evaluated from video recordings of voluntary pursuit movements of the mice (Figs. 2e, f). The mouse, when held by its tail and allowed to grasp a pencil with its fore paws, tries to catch the object with its hind paws and extends simultaneously both hind limbs. In intact animals, the relative length of the two extremities, as estimated by lines connecting the most distal mid-point of the extremity with the anus, is approximately equal and the PLR (ratio of the right to left limb length) is close to 1 (Fig. 2e). After denervation, the limb cannot extend maximally (left side of the animal shown in Fig. 2f) and the PLR increases significantly above 1.

Retrograde labeling of motor neurons

Three months after nerve transection, the animals were anaesthetized with fentanyl, droperidol and diazepam. After exposing the left femoral nerve, a piece of Parafilm (Pechiney Plastic Packaging, Chicago, IL, USA) was laid underneath the nerve trunks distal to the bifurcation, and the two nerve branches were transected 5 mm distal to the bifurcation. Fluorescence retrograde tracers were applied to the cut nerve ends in the form of powder: Fluoro-Ruby (tetramethylrhodamine dextran, Molecular Probes, Eugene, Oregon, USA) to the muscle branch, Fast-Blue (EMS-Chemie GmbH, Großumstadt, Germany) to the cutaneous branch. Thirty minutes after dye application, the nerve stumps were rinsed thoroughly with PBS and gently blotted with filter paper. The wound was surgically closed. Seven days later, the mice were deeply anaesthetized with sodium pentobarbital and transcardially perfused with PBS for 1 min followed by 4% formaldehyde in 0.1 M cacodylate buffer, pH 7.4, for 20 min. The lumbar spinal cord was removed, postfixed overnight at 4°C in the same fixative and cut transversely (serial section of 50 μm thickness) on a Leica VT 1000 M Vibratome (Leica). The sections were coverslipped with Fluoromount-G (SouthernBiotech, Birmingham, AL, USA) and examined under an Axiophot 2 microscope (Zeiss) with filter sets appropriate for Fluoro-Ruby and Fast-Blue. About 45–50 serial cross-sections contained retrogradely labeled cells. All sections were examined and all single- and double-labeled cells were counted except for those profiles visible on top of sections. The application of this simple stereological principle prevents double counting of labeled cells and allows an unbiased evaluation of cell numbers which does not rely on assumptions or require corrections.

Histological analysis of regenerated nerves

The common femoral nerve trunk was removed with both branches, postfixed in osmium tetroxide and embedded in Epon according to a routine protocol. Semithin (1 μm) transverse sections were cut from the nerve branches on a Reichert Ultracut UCT (Leica) and stained with 1% Toluidine blue/1% borax. The number of myelinated axons in the muscle and cutaneous nerve branches was counted on an Axioskop microscope (Zeiss) equipped with a motorized stage and Neurolucida software-controlled computer system (MicroBrightField Europe, Magdeburg, Germany) using a 100× objective.

Statistical analyses

Data were analyzed using appropriate parametric or non-parametric tests as indicated in the text and figure legends. The accepted level of significance was 5%. Throughout the text and in the figures data are presented as mean values ± standard error of mean (SEM).

Results

In intact mice, the HNK-1 carbohydrate is detectable by indirect immunofluorescence in myelin sheaths of large-diameter axons in the femoral nerve prior to its bifurcation and in the motor branch, but barely in the sensory branch beyond the bifurcation point (Martini et al., 1992, 1994). One to three weeks after nerve transection, a similar staining pattern is observed in the proximal nerve stump in which the structural integrity of axons and myelin is largely preserved (Fig. 3b, compare with the negative staining control shown in Fig. 3a). In the distal nerve branches, the staining pattern is changed as a result of axon and myelin degeneration (Wallerian degeneration) and subsequent Schwann cell proliferation and axonal regrowth from the proximal nerve stump (compare the motor nerve branch on the right-hand side in Fig. 3c with the proximal nerve stump shown in Fig. 3b). Note higher expression of the HNK-1 epitope in the motor branch compared to the sensory nerve branch following nerve transection (Fig. 3c). The staining pattern and intensity in electrically stimulated nerves (Fig. 3d) appeared to be altered as compared to sham-stimulated nerves (Fig. 3c). To verify the existence of differences, we evaluated quantitatively staining intensities and counted numbers of HNK-1 positive structures in the regenerating nerve branches at 1, 2 and 3 weeks after nerve injury.

HNK-1 expression in electrically stimulated and sham-stimulated nerves of C57BL/6J mice

In the quadriceps branch, the density of HNK-1 positive structures 1 to 3 weeks after nerve transection was significantly lower in sham-stimulated nerves (12 mice per time-point studied) compared to intact contralateral nerves (Fig. 4a). This injury-induced decline in HNK-1 immunoreactive structures was not found in electrically stimulated nerves (12 mice per time-point studied) in which the numbers were similar to control

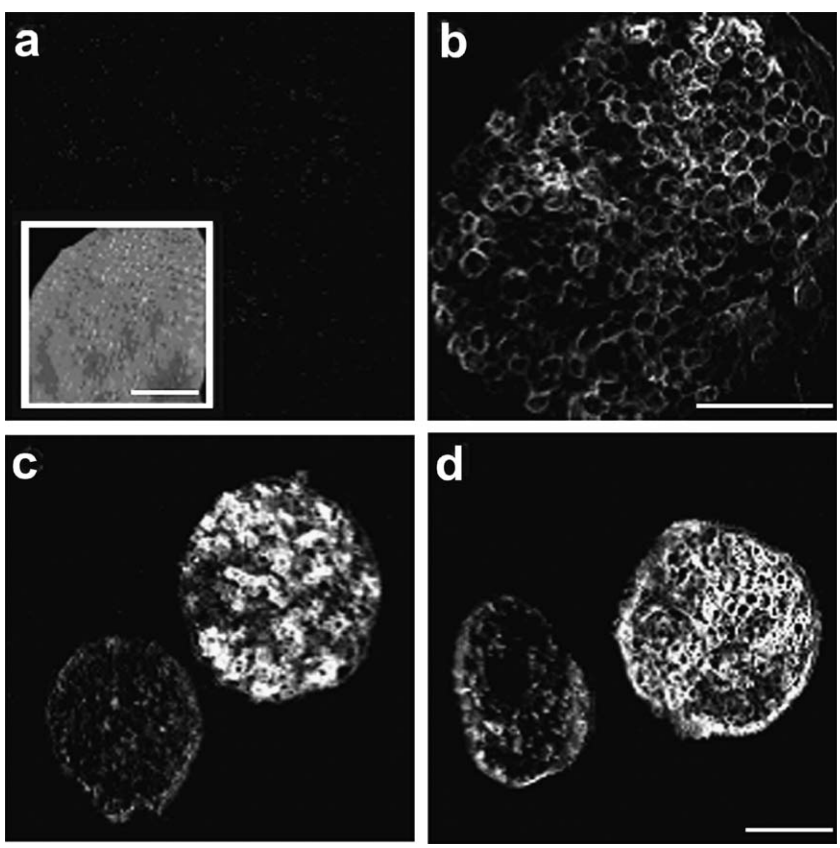

Fig. 3. HNK-1 expression in the femoral nerve and its branches. Shown are cross-sections from the femoral nerve proximal to the transection site (a, b) and immediately distal to the bifurcation (c, d) of a sham-stimulated (a–c) and an electrically stimulated animal (d) studied 1 week after surgery. The quadriceps branch is seen on the right hand side in panels c and d. Panel a shows a negative control section (omission of primary antibody) and a phase-contrast view (insert) of the section observed with epi-fluorescence in panel a. Scale bars shown in the insert, in panels b and d, indicate 100 μm for the insert, panels a, b and c, d, respectively.

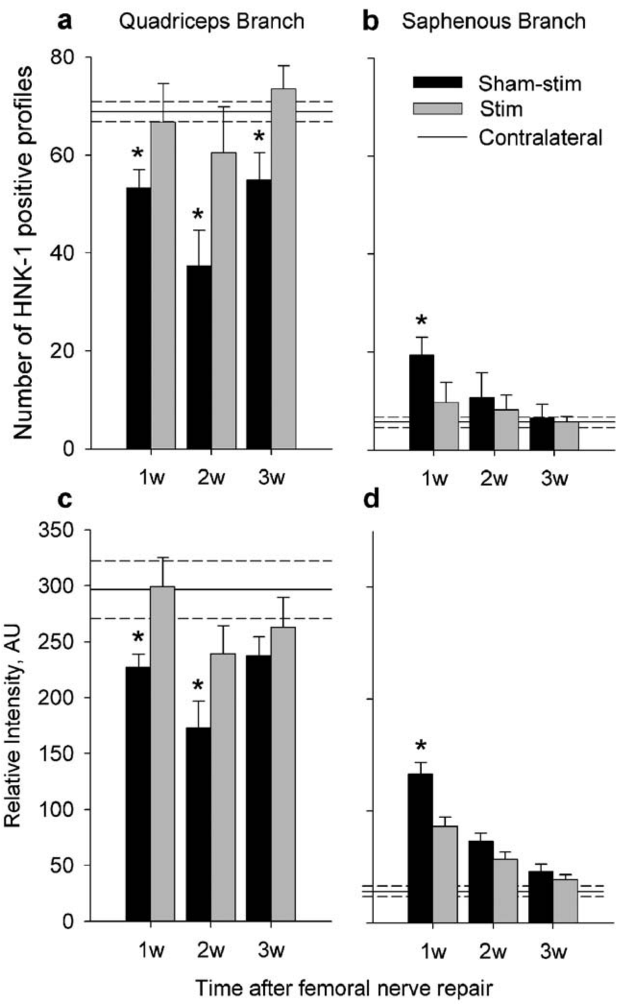

Fig. 4. Numbers of HNK-1 positive profiles (a, b) and relative intensity of HNK-1 immunofluorescence (c, d) in the quadriceps (a, c) and in the saphenous (b, d) branches of femoral nerves studied 1 to 3 weeks after nerve transection followed by electrical stimulation (black columns) or sham-stimulation (white columns). The values for the contralateral untreated nerves are shown as horizontal lines. Asterisks indicate significant differences between stimulated and non-stimulated nerves at a given time-period ($P < 0.05$, non-parametric Kruskal–Wallis analysis of variance, ANOVA, with Dunn's post hoc test). In the same tests, the sham-stimulated groups for the quadriceps branch at 1 to 3 weeks, and the saphenous branch at 1 week differed from contralateral control nerve values shown in panels a and b. For the data shown in panels c and d, the sham-stimulated groups for the quadriceps branch at 2 weeks and both experimental groups for the saphenous branch at 1 and 2 weeks differed from contralateral control nerve values.

values and significantly higher compared to sham-stimulated nerves (Fig. 4a). Measurements of the relative fluorescence intensities yielded similar results (Fig. 4c).

Femoral nerve injury and sham stimulation induced enhanced HNK-1 expression in the saphenous branch, contrary to the quadriceps branch in the same mice, at 1 week after surgery as compared to intact nerves which gradually declined to control levels at 3 weeks (Figs. 4b, d). Similar to the motor branch, however, electrical stimulation counteracted the injury-induced changes and significantly reduced the up-regulation of HNK-1 expression at 1 week after surgery (Figs. 4b, d).

These results show that nerve damage causes opposite changes in the two femoral nerve branches: a decrease in the high expression level in the quadriceps branch and an increase of the normally low HNK-1 expression in the saphenous branch. The injury-induced reduction in the chemical polarization of the two nerve branches is counteracted by electrical stimulation. It should be noted that, without electrical stimulation, the HNK-1 expression level in the motor nerve branch 1 week after lesion is reduced by some 25% only compared to intact nerves (Figs. 4a, c). The expression level is still 2- to 3-fold higher compared with the saphenous branch at 1 week and the ratio increases thereafter in parallel to the previously documented gradual increase in preferential motor reinnervation.

Effects of electrical stimulation on HNK-1 expression in TrkB- and BDNF-deficient mice

We studied the expression of the HNK-1 epitope in mice heterozygous and thus partially deficient in the expression of BDNF or its receptor TrkB. Without electrical stimulation, reduction in HNK-1 expression in the quadriceps branch and increase of this expression in the saphenous branch were found at 2 and 3 weeks after surgery in both BDNF (4 mice per time-point studied) and TrkB-deficient mice (7 per group), as well as in a group of simultaneously treated wild-type littermates (8 per group) (Figs. 5a, b). In this latter group of control animals, we again found that electrical stimulation significantly increased HNK-1 expression in the quadriceps branch and decreased this expression in the saphenous branch (Figs. 5a, b). The effects of electrical stimulation in both BDNF (5 mice per group) and TrkB-deficient mice (7 per group) were, however, not significant indicating that alterations in HNK-1 expression induced by electrical stimulation are strongly dependent on TrkB signaling.

Preferential motor reinnervation in TrkB-deficient mice and wild-type littermates

If TrkB signaling and HNK-1 expression are causally related to preferential motor reinnervation, one should expect that reduced levels of TrkB expression will compromise selectivity of reinnervation. To test this hypothesis, we performed double retrograde labeling of motoneurons that had reinnervated either the saphenous or the quadriceps branch only or both branches 3 months after nerve surgery. In TrkB-deficient mice (TrkB+/−, $n = 8$, Fig. 6a), equal numbers of motoneurons had innervated either the motor or sensory branch, and a minority had innervated both branches. While numbers of motoneurons labeled via the sensory branch alone or via both nerve branches were similar in wild-type mice (TrkB+/+, $n = 8$, Fig. 6a) and TrkB+/− littermates, neurons innervating the motor branch were significantly more numerous in TrkB+/+ versus TrkB+/− mice ($P < 0.05$, t test). These findings show a preferential motor reinnervation of the quadriceps nerve branch in TrkB+/+ mice (index of preferential motor reinnervation, PMR, bar pair to the

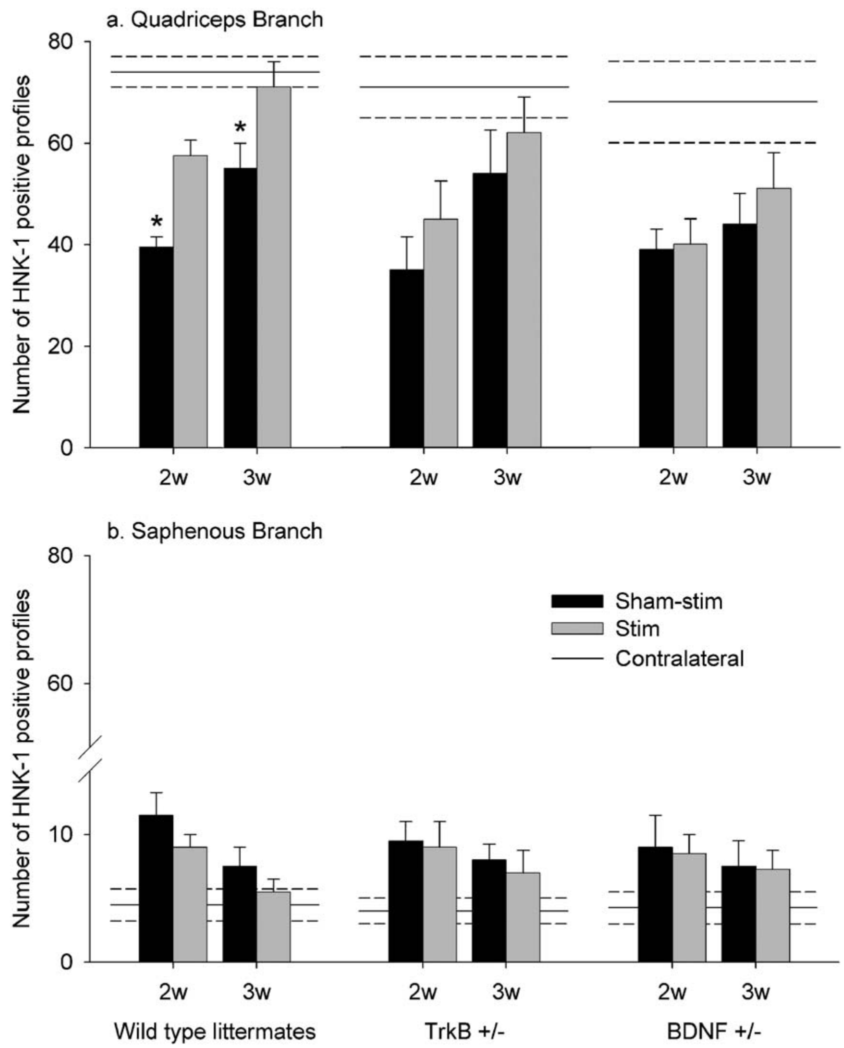

Fig. 5. Numbers of HNK-1 positive profiles in the quadriceps (a) and in the saphenous (b) branches of femoral nerves of wild-type animals (group of bars on the left-hand side), TrkB+/− mice (group of bars in the middle) and BDNF+/− mice (group of bars to the right). The animals were studied 2 and 3 weeks after nerve transection followed by electrical stimulation (black columns) or sham-stimulation (white columns). The values for the contralateral untreated nerves are shown as horizontal lines. Asterisks indicate significant differences between stimulated and non-stimulated nerves at given time-period ($P < 0.05$, non-parametric Kruskal–Wallis analysis of variance, ANOVA, with Dunn's post hoc test). In the same test, the stimulated and sham-stimulated groups for both branches at 2 weeks differed from contralateral control nerve values.

right-hand side in Fig. 6a), but not in TrkB+/− mice (Fig. 6a, $P < 0.05$ compared to TrkB+/+ littermates, *t* test).

To estimate the number of motoneurons normally innervating the quadriceps muscle in TrkB+/+ and TrkB+/− mice, we transected the femoral nerve in 4 animals of each genotype, applied tracer and perfused the animals 1 week later. No difference in the size of the quadriceps motoneuron pool was found between TrkB+/− mice (184 ± 13 motoneurons) and TrkB+/+ littermates (187 ± 11, not significant, *t* test). Compared with these control values, the total number of retrogradely labeled motoneurons after a regeneration period of 3 months (Fig. 6a) was reduced by 30% and 20% in TrkB+/− and TrkB+/+ mice, respectively ($P < 0.05$ for both genotypes, *t* test). Finally, we should note that the total number of motoneurons that had regenerated their axons into the motor nerve branch, i.e. neurons labeled through the motor branch plus double-labeled neurons, was significantly lower in the TrkB+/− mice (71 ± 6) versus TrkB+/+ mice (93 ± 9, $P < 0.05$, *t* test). These numbers represent only 39% and 50%, respectively, of the motoneurons which normally innervate the quadriceps muscle (around 180 motoneurons found in control animals of both genotypes, see above), a finding important with regard to the degree of functional recovery achieved after nerve injury (see below).

Numbers of myelinated axons in regenerated nerves in TrkB-deficient and wild-type littermate mice

One month after nerve transection, numbers of myelinated axons in both motor and sensory nerve branches were lower (by 33% and 24%, respectively) in TrkB+/− versus TrkB+/+ littermates (Fig. 6b, $P < 0.05$ for both branches, *t* test, 8 mice per genotype). While the values for TrkB+/+ mice did not change between 1 and 3 months of reinnervation (differences of

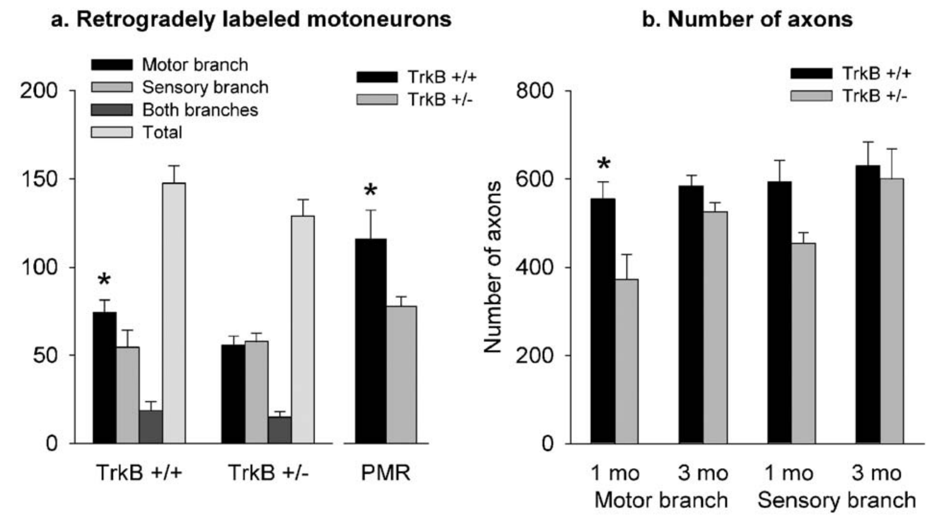

Fig. 6. Morphological evaluation of muscle reinnervation (a) and nerve regeneration (b) after nerve transection in TrkB+/+ and TrkB+/− mice. The two groups of bars on the left hand side in panel a show numbers of motoneurons innervating the motor (M) or sensory (S) branches only, or both branches (B), as well as the total number of regenerated motoneurons (T) 3 months after nerve lesion. The number of motoneurons with axons regenerated into the motor branch only (M, black bars) is significantly higher in TrkB+/+ ($n = 74 \pm 7$) compared to TrkB+/− ($n = 56 \pm 5$, asterisk, $P < 0.05$, t test) mice while no significant differences were found for the other categories of regenerated neurons (t test). The degree of preferential motor reinnervation (bar pair on the right hand side, values multiplied by 100 to fit the scaling), calculated as ratio of the number of motoneurons regenerated into the motor branch only (M) to those regenerated into the sensory branch (S), is also significantly higher in TrkB+/+ animals (dark gray bar) compared to TrkB+/− (gray bar, $P < 0.05$, t test) mice. (b) Numbers of myelinated axonal profiles in semithin sections from the motor and the sensory nerve branches 1 and 3 months after nerve transection in TrkB+/+ ($n = 555 \pm 37$ and 584 ± 24, respectively) and TrkB+/− mice ($n = 373 \pm 56$/534 ± 22, respectively). The asterisk indicates significant difference between the mean group values ($P < 0.05$, t test).

about 5% for both nerve branches, Fig. 6b), numbers of myelinated axons in TrkB+/− nerves increased to reach near-control levels at 3 months (Fig. 6b, 8 mice per genotype). Therefore, partial deficiency in TrkB expression causes a significant delay in axonal regrowth and myelination.

Recovery of motor function in TrkB-deficient mice and wild-type littermates

The finding of delayed and less precise reinnervation in the TrkB+/− mice raised the question as to whether also motor function recovery is delayed and/or less complete in TrkB+/− compared to TrkB+/+ mice. Using a novel single-frame motion analysis approach, we have recently provided evidence that TrkB deficiency causes impairment of functional recovery at early time-periods after nerve injury (Irintchev et al., 2005). To confirm this result and extend the observations to long-term recovery periods, we evaluated the extensor function of the quadriceps muscle in mice of both genotypes (16 mice per group) prior to nerve transection, and at 1, 2, 4, 8 and 12 weeks after surgery (Figs. 7a–c). Three parameters were used for evaluation, the foot-base angle (FBA, Figs. 2a, b, 7a), the heels-tail angle (HTA, Figs. 2b, c, 7b) and the limb protraction length ratio (PLR, Figs. 2e, f, 7c). The first two parameters, evaluated from video recordings of beam walking trials, reflect the abilities of the quadriceps muscle to keep the knee extended during single support phases of the gate cycle and, for the FBA, to counteract in addition the rotational moments in the hip caused by the swing of the contralateral extremity. These abilities were severely compromised after femoral nerve injury as seen from the changes in the angle values at 7 days after nerve transection compared to the preoperative values (day 0) of the same animals (Figs. 7a, b). Multifactor analysis of variance (MANOVA) for repeated measurements showed that both genotypes (TrkB+/+ versus TrkB+/−) and recovery period have a strong impact on the two parameters, with a significant interaction between the two factors. This shows that functional recovery is affected by reduced TrkB expression. As indicated by multiple comparisons between mean group values using the stringent post hoc test of Tukey, the most pronounced difference was observed at 4 weeks after lesion, when motor function recovery in TrkB+/− animals was by far less than in TrkB+/+ littermates (Figs. 7a, b, asterisks). The large difference between the two groups at 4 weeks is even more apparent for the stance recovery index (Fig. 7d, see Material and methods for calculation of the index). Functional improvement at 3 months after injury in both groups of animals (Fig. 7d) is remarkable in view of the fact that by that time the quadriceps muscle is reinnervated by only 50% and 39% of the intact motoneuron pool in TrkB+/+ and TrkB+/− mice, respectively (see results of retrograde tracing experiments above). The high degree of recovery despite low numbers of motoneurons is likely to be associated with enlargement of reinnervated motor units that compensates for their reduced numbers (Rafuse and Gordon, 1996). There was a tendency for less complete recovery of FBA in TrkB+/− versus TrkB+/+ littermates.

In addition to the FBA and HTA, which evaluate quadriceps function during ground locomotion, we studied a third functional parameter, the limb protraction length ratio (PLR). This parameter estimates the ability to stretch the knee joint during target-reaching movements performed without body weight support. Denervation-induced functional impairment, which is similar in both genotypes, was found also for these

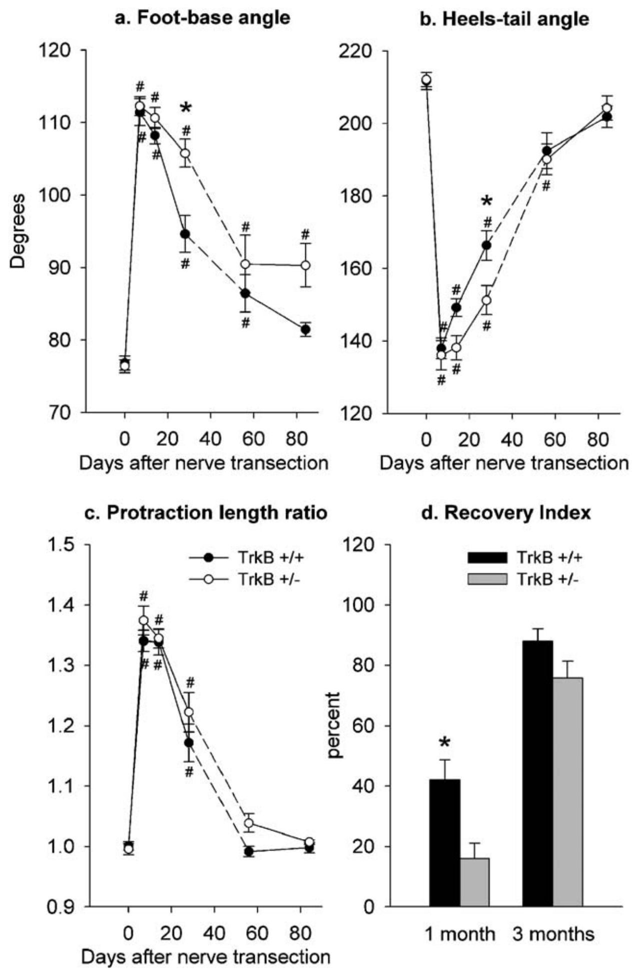

Fig. 7. Functional evaluation of quadriceps muscle reinnervation. Functional performance of TrkB+/+ (open circles and gray bars) and TrkB+/− mice (black circles and bars) was evaluated by measuring foot-base angle (FBA, a), heels-tail angle (HTA, b) and limb protraction length ratio (PLR, c) before lesion (day 0) and during 3 months after nerve transection. Ten TrkB+/+ and nine TrkB+/− animals were studied over a period of 1 month, additional 6 mice per genotype were followed up to 3 months after nerve transection. Statistically significant differences by post hoc Turkey's test analysis performed after multifactor analysis of variance (MANOVA) for repeated measurements are also indicated for group mean values different from: (#) the within-group preoperative value or (*) from both the within-group preoperative value and the corresponding postoperative value of the other group. (d) Stance recovery index, an overall estimate of quadriceps muscle function during stance, at 1 and 3 months after nerve transection in the TrkB+/+ (black bars) and TrkB+/− (gray bars) mice shown in panels a–c. The asterisk shown in panel d indicates a statistically significant difference between the two groups ($P < 0.05$, t test).

parameters (Fig. 7c, compare values at 0 and 7 days). As indicated by MANOVA analysis, PLR was significantly affected by recovery period but not by genotype. Thus, recovery of the ability to use the limb without bearing body weight was not compromised in TrkB+/− versus TrkB+/+ littermates.

Discussion

We have previously shown that the HNK-1 carbohydrate is specifically expressed in motor, but not sensory, axon pathways and proposed that this expression contributes to the preferential motor reinnervation of the quadriceps as opposed to the saphenous branch of the femoral nerve in rodents (Martini et al., 1992, 1994). Recent work has provided direct evidence for the importance of the HNK-1 carbohydrate in motor reinnervation (O. Simova, A. Irintchev, M. Schachner and colleagues, unpublished observations). Application of a HNK-1 peptide mimic to the transected femoral nerve of wild-type mice enhanced preferential motor innervation, HNK-1 carbohydrate expression in the motor nerve branch, survival of quadriceps motoneurons and functional recovery compared to three control groups. In accord with this findings, preliminary results from studies on mice deficient in the expression of the HNK-1 sulfotransferase indicate that HNK-1 deficiency causes enhanced loss of motoneurons after facial nerve injury (D.N. Angelov, O. Guntinas-Lichius, A. Irintchev and M. Schachner, unpublished observations). Here, we show that the expression of the HNK-1 epitope is strongly enhanced by short-term electrical stimulation applied to the nerve immediately after transection, a procedure that also leads to improved reinnervation (Al Majed et al., 2000a). Furthermore, we demonstrate that the stimulation-induced HNK-1 up-regulation is dependent on BDNF and its receptor TrkB. Finally, we show that even partially reduced expression of TrkB leads to compromised preferential motor reinnervation, delayed axonal regrowth into the distal nerve stump and delayed functional recovery after femoral nerve injury. In conjunction with previous data, these results show that TrkB signaling is a major factor regulating the functional outcome after nerve injury and that the HNK-1 carbohydrate plays an essential role in the TrkB signaling pathway.

The analysis of HNK-1 expression was based on quantitative immunohistochemistry. Our previous experience showed that immunochemical methods are not applicable for quantification of HNK-1 expression levels. The HNK-1 epitope is carried by both glycoproteins and glycolipids in the femoral nerve branches (Löw et al., 1994) and this makes extraction of small amounts of material that can be obtained from the peripheral nerves from one mouse technically impossible. To verify the reliability of the estimates, we simultaneously used two approaches, counts of immunoreactive profiles and measurements of relative intensities, which produced similar results. In addition, counting of profiles in two different experimental sets of animals indicated the reliability of our measurements. Thus, our data provide sound evidence that electrical stimulation strongly enhances HNK-1 carbohydrate expression in the motor nerve stump and reduces its posttraumatic up-regulation in the sensory nerve branch. This result is remarkable since we stimulated the proximal nerve stump after nerve transection and observed effects in the distal nerve stumps. We cannot rule out the possibility that, in a small animal like the mouse, current from the proximally located stimulation electrode may indirectly, via the surrounding tissue, reach the nerve trunk distal to the transection. Such stimulation of the distal nerve segment may have direct influences early after transection, such as enhancement of Wallerian degeneration and subsequent faster axonal regrowth. Yet, we may propose that the effects of electrical stimulation are dependent on action potential firing and propagated electrical activity in the motoneuron cell bodies and not simply local depolarization of cells in the nerve trunk as shown in rats (Al Majed et al., 2000a). Regrowth of motor axons leads to contact of axons with Schwann cells coincident with up-regulation of HNK-1 immunoreactivity in lesioned nerves (Martini et al., 1992, 1994). It is therefore possible that accelerated up-regulation of HNK-1 expression is the consequence of an accelerated outgrowth of axons from the proximal nerve stump and temporarily earlier contacts of the regrowing axons with Schwann cells (Brushart et al., 2002). Motor axons are likely to induce HNK-1 expression in Schwann cells by direct surface contact or by supply of growth factors from the growth cone. Also, electrical stimulation may improve the capacity of regenerating axons to induce changes favoring regeneration in the denervated nerve environment. In accordance, enhanced expression of both BDNF and TrkB in electrically stimulated motoneurons (Al Majed et al., 2000b) has been found. Enhanced expression of BDNF might induce Schwann cells to up-regulate HNK-1 expression in the quadriceps branch and reduce it in the saphenous branch, via paracrine mechanisms involving the secretion of BDNF from regenerating axons. This functionally meaningful opposite regulation of HNK-1 expression in the motor and sensory pathways might be possible because the Schwann cells that originally have been in contact with motor versus sensory axons respond differently to regeneration-related cues of motor and sensory axons (Martini et al., 1994). Schwann cells during development and after a lesion express different neurotrophin receptors, including the low affinity p75 receptor and the truncated TrkB receptor (Ernfors et al., 1989; Funakoshi et al., 1993). Differential expression of receptors in the two nerve branches and/or different signaling pathways in motor versus sensory axon-associated Schwann cells may explain the opposite effects seen in the motor and sensory nerve branches.

Neurotrophins like NGF have been shown to influence expression of adhesion molecules, such as the potent neurite outgrowth promoting molecule L1 in neurons (Lee et al., 1981). Up-regulation of neurotrophin expression in neurons and possibly in Schwann cells may be essential for regulating speed of axonal regrowth. It is interesting in this respect that regulation of L1 synthesis can be affected by both neurotrophins and electrical stimulation (Seilheimer and Schachner, 1987; Scherer et al., 1992; Itoh et al., 1995, 1997). The fact that L1 expression is conducive to axon growth when expressed and up-regulated not only in neurons, but also in glial cells of the peripheral and central nervous system (Mohajeri et al., 1996; Yazaki et al., 1996) points to concerted actions of neurotrophins and electrical activity in nerve regeneration: generally expressed adhesion molecules provide the overall conducive environment for axon growth, glycans carried by these molecules or by glycolipids then provide the fine tuning of pathway choices at points of target decision. Recent findings indicate that a glycan, polysialic acid, attached to the cell adhesion molecule NCAM, a HNK-1 binding partner (Cole and Schachner, 1987), is essential for selective targeting of regenerating motoneurons in the mouse femoral nerve paradigm (Franz et al., 2005). These observations further strengthen the importance of recognition molecules and, in particular their associated glylcans, as regulators of peripheral nerve regeneration. It will be important to further investigate the molecular mechanisms promoting regeneration and functional recovery in the peripheral nervous system via growth- and survival-conducive adhesion molecules and associated glycans, since these could provide guides to understanding and manipulating recovery of function in patients after peripheral nerve lesions.

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